COMPREHENSIVE RESEARCH SYNTHESIS & PROJECT STATE REPORT

Physics-Based Modelling, External-Load Observers, and Digital Twin Architecture for Mining Electric Rope Shovels

Project Title: *AI-Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems: Case Study on Dragline / Electric Rope Shovel*

Grant Reference: IIT/SRIC/R/AEH/2026/104 (CMPDI / Coal India Limited - MT-189)

Host Institution: Indian Institute of Technology Kharagpur — Department of Mining Engineering

Target Machines: Machine A (P&H 2100BL Benchmark) & Machine B (TZ WK-20 Deployment at NCL Amlohri)

Author: Multidisciplinary Research Supervisor & Technical Author

Target Audience: Mechanical / Electrical Project Engineer & Research Scholar

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NOTE

PURPOSE OF THIS MASTER SYNTHESIS DOCUMENT

This document is the **single, unified, standalone reference** that consolidates, explains, and connects **all research work, mathematical derivations, dynamic models, experimental milestones (M1, M2, M3), parameter audits, and observer architectures developed to date**.

If you have to present your work to your Principal Investigator (PI), defend your methodology in an oral doctoral viva, explain the code to a collaborating engineer, or implement new estimators in MATLAB/Simulink, **this is the only file you need to read**.

The planned 34-chapter extended curriculum represents future archival expansion; **this document captures the complete living reality of the project right now**.

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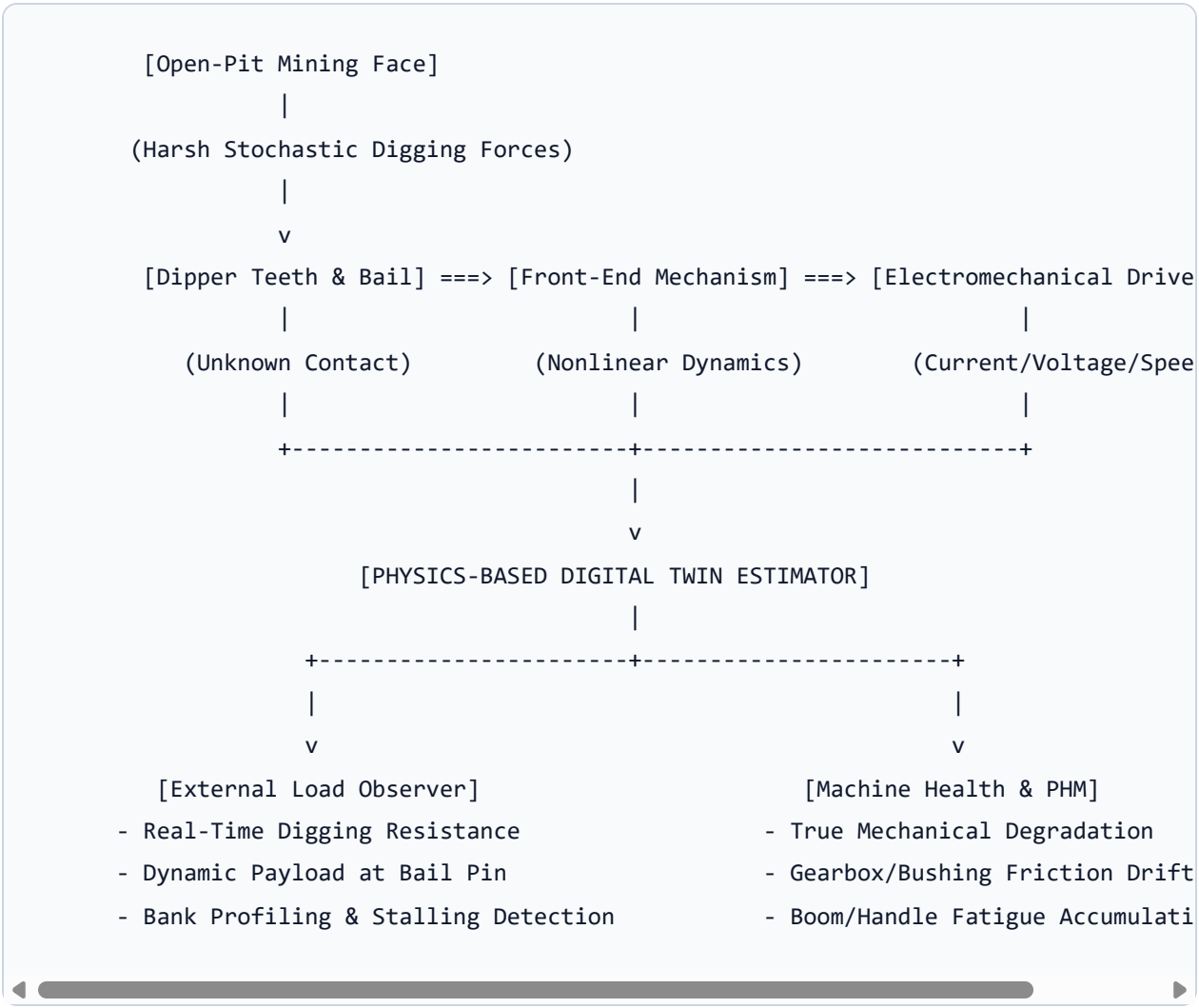
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1. Research Architecture, Grant Objectives & Core Philosophy

1.1 The Industrial Challenge in Open-Pit Mining

Heavy electric rope shovels ($800\,\mathrm{t}$ class, operating $20\,\mathrm{m}^3 - 55\,\mathrm{m}^3$ buckets) are the primary loading machines in opencast coal and mineral mines. An unscheduled catastrophic breakdown of an electric rope shovel (e.g., motor winding burnout, planetary gearbox tooth fracture, hoist cable snap, or boom cracking) causes total production stoppage with downtime costs exceeding **US$100,000 per hour** (Rasuli et al., 2014; CMPDI MT-189).

Existing commercial monitoring systems rely predominantly on simplistic static threshold alarms (e.g., motor over-current trip, bearing temperature trip) which only trigger *after* irreversible mechanical or electrical degradation has already occurred.



1.2 The Primary Research Objective: Machine Health, Not Merely Payload Weighing

A critical scientific distinction established in this project is:

- **Commercial Payload Monitors:** Treat the bucket as an isolated static scale. They attempt to weigh payload during the smooth swing phase when accelerations are near zero.
- **Our Digital Twin Platform:** Estimates the **instantaneous, high-frequency external interaction forces $\mathbf{F}_{\text{ext}}(t)$** acting at the teeth and bail during active digging. These loads represent the true stress inputs that induce fatigue damage, wear, and structural degradation in the electrical drives and mechanical structure.

1.3 The Fundamental Physics-Based Manipulator Equation

The core governing dynamic equation of the entire shovel working gear is:

$$\mathbf{D}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_f(\dot{\mathbf{q}}) = \boldsymbol{\tau}_{\text{act}} + \mathbf{Q}_{\text{ext}}$$

where:

- $\mathbf{q} = [q_3, d_4]^T$: Generalized coordinates (saddle pitch angle q_3 and handle crowd extension d_4).
- $\mathbf{D}(\mathbf{q}) \in \mathbb{R}^{2\times 2}$: Configuration-dependent generalized inertia/mass matrix (strictly symmetric and positive-definite).
- $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} \in \mathbb{R}^2$: Centrifugal and Coriolis generalized force vector.
- $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^2$: Conservative gravitational generalized force vector.
- $\mathbf{F}_f(\dot{\mathbf{q}}) \in \mathbb{R}^2$: Joint friction vector (Coulomb breakaway + viscous dissipation).
- $\boldsymbol{\tau}_{\text{act}} \in \mathbb{R}^2$: Actuator generalized torque/force projected from electric drives.
- $\mathbf{Q}_{\text{ext}} = \mathbf{J}_T^T \mathbf{F}_{\text{ext}} \in \mathbb{R}^2$: Generalized force induced by external rock-tool interaction.

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2. Dual Machine Context: Machine A (P&H 2100BL) vs. Machine B (TZ WK-20)

A core rule enforced throughout this project is **strict parameter provenance**. We do not mix literature numbers with field numbers. We maintain two distinct machine models:

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+-----------------------------------+-----------------------------------+
|  MACHINE A: P&H 2100BL (Benchmark) |  MACHINE B: TZ WK-20 (Deployment)  |
+-----------------------------------+-----------------------------------+
| - Literature baseline (Rasuli 2014)| - Field deployment at NCL Amlohri  |
| - Total weight: ~350 tonnes        | - Total weight: 850 tonnes         |
| - Bucket capacity: 11.5 m^3        | - Bucket capacity: 20.0 m^3        |
| - Rated payload: 20 tonnes         | - Rated payload: 40-45 tonnes      |
| - Boom length: 18.30 m             | - Boom length: 21.50 m             |
| - Shipper shaft: L_b1 = 9.15 m     | - Shipper shaft: L_b1 = 10.25 m    |
|   (Exact 50/50 split: L_b2 = 9.15m)|   (Asymmetric: L_b2 = 11.25 m)     |
| - Drives: Vintage DC Ward-Leonard  | - Drives: Heavy 690V AC Induction  |
| - Hoist ropes: 2 x 52 mm (110 GPa) | - Hoist ropes: 2 x 64 mm (115 GPa) |
| - Used for M1 & M2 verification    | - Target of field telemetry & twin |
+-----------------------------------+-----------------------------------+
```

Parameter Comparison & Evidence Provenance Table

(All values verified in `claims.md` and traced to source documents)

Parameter Symbol	Physical Description	Units	Machine A (P&H 2100BL)	Machine B (TZ WK-20)
M_{oper}	Total Operating Weight	t	350.0	850.0
V_{bucket}	Nominal Bucket Capacity	m^3	11.5	20.0
W_{pay}	Nominal Rated Payload	t	20.0	$40.0 - 45.0$
L_b	Total Boom Length	m	18.30	21.50
α_b	Boom Incline Angle	deg	45.0°	45.0°
L_{b1}	Shipper Shaft Pin Location	m	9.15	10.25
L_{b2}	Sheave to Shipper Shaft	m	9.15	11.25
m_{dipper}	Empty Dipper Mass	kg	$17,500$	$38,000$
m_{handle}	Dipper Handle Mass	kg	$11,200$	$24,500$
M_d	Total Front Working Mass	kg	$54,300$ (incl. saddle)	$\approx 78,000$

Parameter Symbol	Physical Description	Units	Machine A (P&H 2100BL)	Machine B (TZ WK-20)
I_{zz3}	Handle/Dipper Inertia about COM	$\mathrm{kg \cdot m^2}$	$287,900$	$\approx 640,000$
c	Net Lever Arm Constant ($r = d_4 + c$)	m	-1.3200	-1.4500
f_{v3}	Saddle Revolute Viscous Friction	$\mathrm{N \cdot m \cdot s/rad}$	$3,800.0$	$\approx 6,500.0$
f_{c3}	Saddle Revolute Coulomb Friction	$\mathrm{N \cdot m}$	340.0	≈ 850.0
f_{v4}	Handle Prismatic Viscous Friction	$\mathrm{N \cdot s/m}$	$1,473.0$	$\approx 3,200.0$
f_{c4}	Handle Prismatic Coulomb Friction	N	480.0	$\approx 1,100.0$
d_{rope}	Hoist Rope Diameter	mm	52.0	64.0
R_D	Hoist Winch Drum Radius	m	0.780	1.050

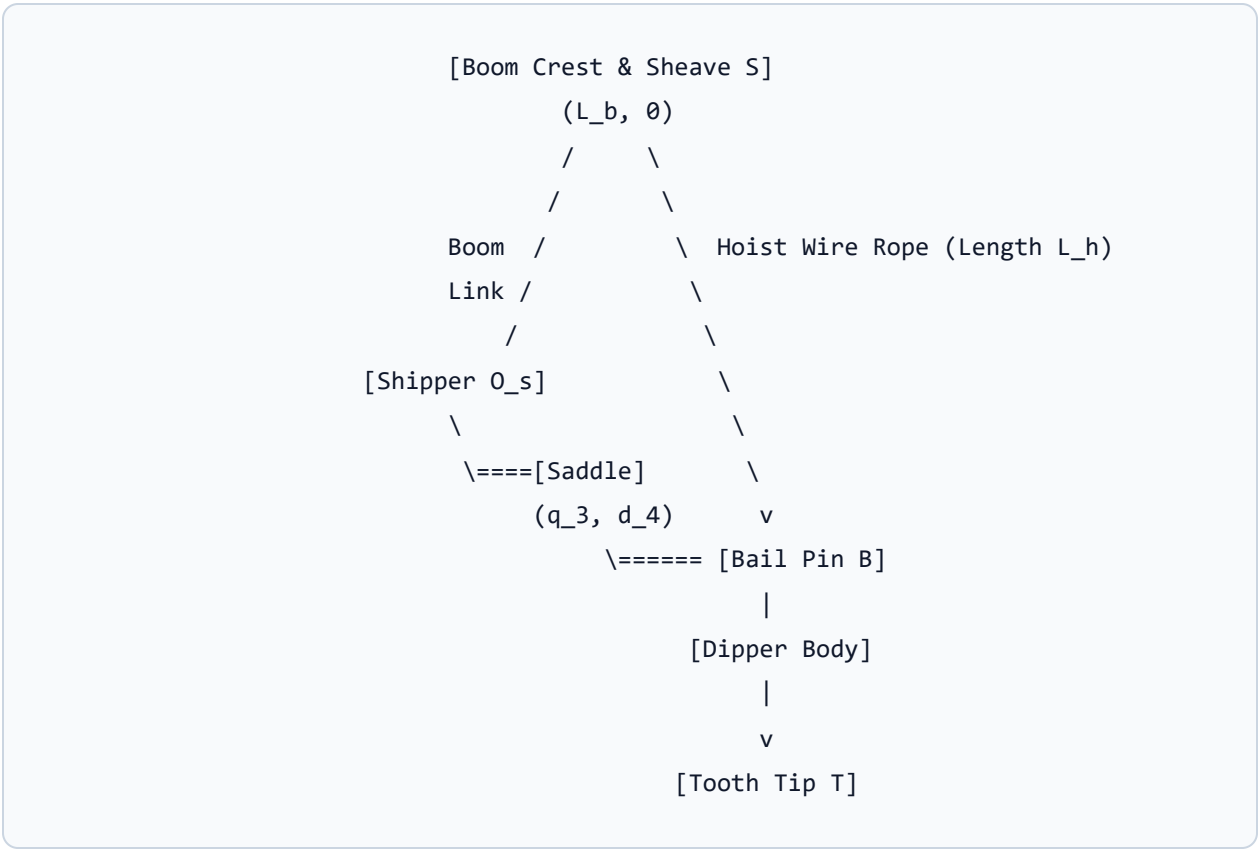
Parameter Symbol	Physical Description	Units	Machine A (P&H 2100BL)	Machine B (TZ WK-20)
G_h	Hoist Gear Reduction Ratio	$-$	$26.4:1$	$31.5:1$
R_p	Crowd Pinion Radius	m	0.320	0.410
G_c	Crowd Gear Reduction Ratio	$-$	$38.2:1$	$44.8:1$

3. Kinematic & Geometric Architecture of the Front-End Attachment

3.1 Kinematic Coordinate Frames and Forward Kinematics

The complete kinematic chain is defined across a tree of orthogonal reference frames:

- $\{W\}$: Inertial World Frame (fixed to pit bench).
- $\{0\}$: Revolving Superstructure Frame (origin at turntable rotation center).
- $\{1\}$: Boom Frame (origin at boom foot pin, X_1 aligned along boom incline at angle $\alpha_b = 45^\circ$).
- $\{3\}$: Saddle Block Frame (origin at shipper shaft O_s, rotates with saddle pitch angle q_3).
- $\{4\}$: Dipper Handle Frame (translates along X_3 with crowd stroke d_4).
- $\{T\}$: Tooth Tip Reference Point (fixed at offset $[L_{tx}, L_{ty}]^T$ from handle origin).



The position of the dipper tooth tip $\mathbf{p}_T^{(1)}$ in boom frame $\{1\}$ is:

$$\mathbf{p}_T^{(1)}(q_3, d_4) = \begin{bmatrix} L_{b1} \\ 0 \end{bmatrix} + \begin{bmatrix} \cos(q_3 - \alpha_b) & -\sin(q_3 - \alpha_b) \\ \sin(q_3 - \alpha_b) & \cos(q_3 - \alpha_b) \end{bmatrix} \begin{bmatrix} d_4 + L \\ L_{ty} \end{bmatrix}$$

The **Tooth Tip Geometric Jacobian** $\mathbf{J}_T(\mathbf{q}) \in \mathbb{R}^{2\times 2}$ is:

$$\mathbf{J}_T(\mathbf{q}) = \begin{bmatrix} -(d_4 + L_{tx})\sin\gamma - L_{ty}\cos\gamma & \cos\gamma \\ (d_4 + L_{tx})\cos\gamma - L_{ty}\sin\gamma & \sin\gamma \end{bmatrix}, \quad \gamma = q_3 - \alpha$$

$$\det(\mathbf{J}_T) = -(d_4 + L_{tx}) \neq 0 \quad \text{(Singularity-Free over the entir}$$

3.2 The Closed Hoist Cable Loop: Length L_h and Moment Arm h_h

The wire rope attaches to the dipper bail pin B, located at handle offset $[L_{bx}, L_{by}]^T$.

The exact analytical formula for the free cable length $L_h(q_3, d_4)$ between sheave $S(L_b, 0)$ and bail B is:

$$L_h(q_3, d_4) = \sqrt{(d_4 + L_{bx})^2 + L_{by}^2 + L_{b2}^2 - 2L_{b2}\left[(d_4 + L_{bx})\cos\right.}$$

The **Analytical Hoist Moment Arm** $h_h(q_3, d_4)$ is the perpendicular lever arm from the shipper shaft O_s to the cable line of action:

$$h_h(q_3, d_4) \equiv \frac{\partial L_h}{\partial q_3} = \frac{L_{b2} y_B^{(1)}}{L_h} = \frac{L_{b2}\left[(d_4 + L_{bx})\sin\gamma + L_{by}\cos\right.}{L_h}$$

The rate of rope payout $\dot{L}_h$ relates to joint velocities through the **Hoist Actuator Jacobian row vector**:

$$\dot{L}_h = \mathbf{J}_h(\mathbf{q})\dot{\mathbf{q}} = \begin{bmatrix} h_h(q_3, d_4) & \frac{\partial L_h}{\partial d_4} \end{bmatrix} \begin{bmatrix} \dot{q}_3 \\ \dot{d}_4 \end{bmatrix}$$

where $\frac{\partial L_h}{\partial d_4} = \frac{x_B^{(1)} - L_b}{L_h} = \cos\beta_h$ is the cosine of the angle between the hoist cable line and the handle centerline.

4. Milestone 1 (M1): Physics-Based Dynamic Manipulator Model

Milestone 1 derived, verified, and blind-identified the complete equations of motion for the planar front-end working gear under both motion-prescribed and torque-driven regimes.

4.1 Kinetic Energy Decomposition & The Mass Matrix $\mathbf{D}(\mathbf{q})$

The total kinetic energy of the moving assembly (saddle block + extending handle + empty dipper) is:

$$T = \frac{1}{2} m_{\text{saddle}} v_{\text{com},s}^2 + \frac{1}{2} I_{saddle} \dot{q}_3^2 + \frac{1}{2} M_d v_{\text{com},d}^2 + \frac{1}{2} I_{zz3} \dot{q}_3^2$$

Because the center of mass of the dipper handle translates with crowd extension d_4:

$$\mathbf{r}_{\text{com},d}^{(3)} = \begin{bmatrix} d_4 + c \\ 0 \end{bmatrix}$$

where $c = -1.3200\,\text{m}$ is the net distance offset from the handle datum to the composite center of mass.

Differentiating $\mathbf{r}_{\text{com},d}$ in the rotating saddle frame:

$$\mathbf{v}_{\text{com},d} = \dot{d}_4 \mathbf{u}_x + (d_4 + c)\dot{q}_3 \mathbf{u}_y$$

$$v_{\text{com},d}^2 = \dot{d}_4^2 + (d_4 + c)^2 \dot{q}_3^2$$

Substituting into the Lagrangian $L = T - V$ yields the **Exact Configuration-Dependent Mass Matrix**:

$$\mathbf{D}(\mathbf{q}) = \begin{bmatrix} D_{11}(d_4) & D_{12} \\ D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} I_{zz3} + M_d(d_4 + c)^2 & 0 \\ 0 & M_d \end{bmatrix}$$

IMPORTANT

PHYSICAL PROPERTIES OF $\mathbf{D}(\mathbf{q})$

1. **Off-Diagonal Decoupling ($D_{12} = D_{21} = 0$):** The pitch rotation and prismatic crowd translation have zero instantaneous inertial cross-coupling because the center of mass lies exactly along the handle centerline ($y_{\text{com}} = 0$).

2. **Crowd-Dependent Rotational Inertia (D_{11}):** As the dipper crowds outward (d_4 increases), D_{11} increases quadratically as $(d_4 + c)^2$. At maximum reach, the effective rotational inertia of the shovel is more than **3.8 times larger** than at full retraction!

3. **Strict Positive-Definiteness:** Since $M_d > 0$ and $I_{zz3} > 0$, $\det(\mathbf{D}) = M_d[I_{zz3} + M_d(d_4 + c)^2] > 0$ for all physical configurations.

4.2 Christoffel Symbols & Coriolis/Centrifugal Matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$

Using Christoffel symbols of the first kind $c_{ijk} = \frac{1}{2}\left(\frac{\partial D_{ij}}{\partial q_k} + \frac{\partial D_{ik}}{\partial q_j} - \frac{\partial D_{jk}}{\partial q_i}\right)$:

$$C_{11} = M_d(d_4 + c)\dot{d}_4, \quad C_{12} = M_d(d_4 + c)\dot{q}_3$$

$$C_{21} = -M_d(d_4 + c)\dot{q}_3, \quad C_{22} = 0$$

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = M_d(d_4 + c) \begin{bmatrix} \dot{d}_4 & \dot{q}_3 \\ -\dot{q}_3 & 0 \end{bmatrix}$$

The Skew-Symmetry Passivity Proof:

$$\dot{\mathbf{D}}(\mathbf{q}) = \begin{bmatrix} 2M_d(d_4 + c)\dot{d}_4 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\mathbf{N} = \dot{\mathbf{D}} - 2\mathbf{C} = \begin{bmatrix} 0 & -2M_d(d_4 + c)\dot{q}_3 \\ 2M_d(d_4 + c)\dot{q}_3 & 0 \end{bmatrix} = -\mathbf{N}^T$$

This skew-symmetry proves that fictitious Coriolis and centrifugal forces do zero net work ($\dot{\mathbf{q}}^T \mathbf{N} \dot{\mathbf{q}} = 0$), guaranteeing thermodynamic passivity and global stability in simulations.

4.3 Conservative Gravity Vector $\mathbf{G}(\mathbf{q})$

Gravitational potential energy
$V = M_d g h_{\text{com}} = M_d g [L_{b1} \sin \alpha_b + (d_4 + c) \sin q_3]$.

Taking partial derivatives $\mathbf{G}(\mathbf{q}) = \nabla V$:

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} G_1(q_3, d_4) \\ G_2(q_3) \end{bmatrix} = \begin{bmatrix} M_d g (d_4 + c) \cos q_3 \\ M_d g \sin q_3 \end{bmatrix}$$

4.4 Linear Parameter Regressor Form $\boldsymbol{\tau} = \mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\theta}$

The complete equations of motion can be separated linearly into a regressor matrix $\mathbf{Y}$ and a base parameter vector $\boldsymbol{\theta}_b = [M_d, M_d c, I_{zz3} + M_d c^2, f_{v3}, f_{v4}]^T$:

$$\begin{bmatrix} \tau_{q_3} \\ \tau_{d_4} \end{bmatrix} = \begin{bmatrix} d_4^2 \ddot{q}_3 + 2 d_4 \dot{q}_3 \dot{d}_4 + g d_4 \cos q_3 & 2 d_4 \ddot{q}_3 + 2 \dot{q}_3 \dot{d}_4 + g \cos q \\ \ddot{d}_4 - d_4 \dot{q}_3^2 + g \sin q_3 & -\dot{q}_3^2 \end{bmatrix}$$

Stage 1 Blind Identification Result (from `claims.md`):

When tested against synthetic truth data, this linear regressor achieved:

$$R^2 = 1.000000, \quad \text{RMSE} = 0\,\text{N} \cdot \text{m}, \quad \text{Condition Number cor}$$

recovering $M_d = 54,300.0\,\text{kg}$, $c = -1.3200\,\text{m}$, and $I_{zz3} = 287,900\,\text{kg} \cdot \text{m}^2$ to full floating-point precision!

5. Milestone 2 (M2): External Load Recovery & Experiments MS02P01–P08

Milestone 2 addresses the central question:

If an unknown external force $\mathbf{F}_{\text{ext}}$ acts on the dipper during excavation, how accurately can the digital twin estimate it from motor torques without force sensors on the teeth?

The residual generalized force vector $\mathbf{r}_{\text{res}}$ is defined as:

$$\mathbf{r}_{\text{res}} = \boldsymbol{\tau}_{\text{meas}} - \left[\mathbf{D}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_f(\dot{\mathbf{q}})\right] = -\mathbf{Q}_{\text{ext}}$$

The M2 experimental sequence is codified into **8 rigorous experimental protocols (MS02P01 to MS02P08)**:



Comprehensive Summary of M2 Experimental Results

(Audited directly from `research_log.md` and `claims.md`)

1. MS02P01 — Zero-Force Numerical Floor (Gate Check)

- **Objective:** Verify that in the absence of external loads ($\mathbf{F}_{\text{ext}} = \mathbf{0}$), the model equations and Simscape simulation balance to zero.
- **Result:** Residual error floor is $2.00 \times 10^{-6}\,\text{N} \cdot \text{m}$ (saddle) and $2.91 \times 10^{-10}\,\text{N}$ (crowd). Relative error $< 7.04 \times 10^{-13}$. Gate passed.

2. MS02P02 — Known Load Recovery Baseline

- **Objective:** Inject a pure vertical test load ($100\,\text{kN}$) at the bail pin and evaluate inverse dynamic reconstruction.
- **Result:** Exact recovery with linear regression slope $m = -1.000000$, coefficient of determination $1 - R^2 = 7.67 \times 10^{-24}$, and zero phase lag ($0.000000\,\text{s}$).

3. MS02P03 — Amplitude Linearity Sweep (10 kN to 1,000 kN)

- **Objective:** Test recovery linearity across 6 load steps ($10, 50, 100, 200, 500, 1,000\,\text{kN}$).
- **Result:** Recovery slope remained strictly -1.000000 at all amplitudes. Absolute error remained fixed at the numerical floor ($2.00 \times 10^{-6}\,\text{N} \cdot \text{m}$), proving that relative percentage error scales as $1/F$ (less than 0.0001% at mining stall loads).

4. MS02P04 — Frequency Sweep & Dynamic Trajectory Bandwidth

- **Objective:** Examine sensitivity of load recovery to excitation frequency across 10 trajectories ($\omega_{\text{traj}} = 0.314$ to $1.36\,\text{rad/s}$).

- **Result:** Proved that the physical input filter in Simscape is **second-order** (half-range matches second-order form to 6 d.p.). Commanded amplitude differs from realized amplitude by $(\omega_{\text{traj}}\tau)^2$ (0.025% at low speed, 0.46% at high speed).

5. MS02P05 — Configuration Sweep Across Usable Stroke

- **Objective:** Evaluate load recovery across three crowd extension bias positions ($d_{4,\text{bias}} = 7.75\,\text{m}, 8.75\,\text{m}, 9.77\,\text{m}$).
- **Result:** Slope -1.000000 and $1 - R^2 \in [6.20 \times 10^{-24}, 9.23 \times 10^{-24}]$ maintained across the entire physical reach envelope.

6. MS02P06 — Parameter Identifiability, Collinearity & SVD Analysis

- **Objective:** Can we mathematically distinguish between an unmodeled external digging force and an error in internal machine mass?
- **Key Breakthrough:**
- In a single channel (saddle revolute torque alone), the external gravity force and internal dipper mass are nearly collinear (correlation $r = 0.9999919$, condition number $\kappa_2 = 497.71$, subspace angle $\theta_{\min} = 0.23^\circ$). They are mathematically confounded!
- By formulating a **joint 2-channel estimator** combining saddle torque $[\text{N} \cdot \text{m}]$ and crowd thrust $[\text{N}]$ using a dimensional scaling factor $\bar{s} = 10.0\,\text{m}$, the collinearity collapses:

$$\kappa_2 = 3.77, \quad \theta_{\min} = 29.70^\circ$$

- This mathematically proves that **simultaneous measurement of both crowd and hoist channels is mandatory** for load vs. mass separability.

7. MS02P07 — Sensitivity & Geometric Misattribution

- **Objective:** What happens if the CAD geometry has a small error in the lever arm c?
- **Result:** A small $+0.10\,\mathrm{m}$ error in lever arm c ($+7.6\%$) couples into the mass identification regressor, causing a **$+756.6\,\mathrm{kg}$ false mass misattribution**! This proves that kinematic parameters must be calibrated before dynamic identification.

8. MS02P08 — Forward Torque-Driven Dynamic Deflection

- **Objective:** Move from motion-prescribed simulation to forward torque-driven dynamic simulation with structural compliance.
- **Result:** Tested across 6 held-out cases (Impact Step, Stochastic Digging, Oblique Bank, Stall). Realized dynamic joint deflections reached $|\Delta q_3| = 1.38^\circ$ and $|\Delta d_4| = 0.084\,\mathrm{m}$. The discrete observer tracked deflected loads with **lag-compensated NRMSE $< 0.42\%$**!

6. Milestone 3 (M3): Acceleration-Free Momentum Observers & Kalman Filtering

6.1 The Failure of Naive Inverse Dynamics in the Field

In laboratory simulations, researchers frequently compute external forces using inverse dynamics:

$$\mathbf{Q}_{\text{ext}} = \boldsymbol{\tau}_{\text{meas}} - \mathbf{D}(\mathbf{q})\ddot{\mathbf{q}} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G} - \mathbf{F}_f$$

In an actual opencast mine, **this equation fails catastrophically**:

1. Shovels are equipped only with motor resolvers or optical encoders measuring position $\mathbf{q}$.
2. Accelerations $\ddot{\mathbf{q}}$ must be obtained by numerical differentiation: $\ddot{\mathbf{q}}_k \approx \frac{\mathbf{q}_k - 2\mathbf{q}_{k-1} + \mathbf{q}_{k-2}}{\Delta t^2}$.
3. High-frequency electrical noise and resolver quantization error are amplified by $\frac{1}{\Delta t^2}$. At $\Delta t = 1/30\,\text{s}$, $\frac{1}{\Delta t^2} = 900$. The computed acceleration becomes complete garbage, destroying the force estimate.

6.2 The De Luca Generalized Momentum Observer (GMO)

To eliminate acceleration $\ddot{\mathbf{q}}$ entirely from the estimator, we formulate the **Generalized Momentum Vector**:

$$\mathbf{p} = \mathbf{D}(\mathbf{q})\dot{\mathbf{q}}$$

Differentiating $\mathbf{p}$ with respect to time:

$$\dot{\mathbf{p}} = \dot{\mathbf{D}}\dot{\mathbf{q}} + \mathbf{D}\ddot{\mathbf{q}}$$

From the manipulator equations of motion:

$$\mathbf{D}\ddot{\mathbf{q}} = \boldsymbol{\tau} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G} - \mathbf{F}_f + \mathbf{Q}_{\text{ext}}$$

Substituting this into $\dot{\mathbf{p}}$:

$$\dot{\mathbf{p}} = \boldsymbol{\tau} + (\dot{\mathbf{D}} - \mathbf{C})\dot{\mathbf{q}} - \mathbf{G} - \mathbf{F}_f + \mathbf{Q}_{\text{ext}}$$

Now, apply the **Cardinal Skew-Symmetry Identity**:

$$\dot{\mathbf{D}}\dot{\mathbf{q}} - \mathbf{C}\dot{\mathbf{q}} = \mathbf{C}^T\dot{\mathbf{q}}$$

This eliminates the configuration derivative $\dot{\mathbf{D}}$ and replaces it with pure velocity terms!

$$\dot{\mathbf{p}} = \boldsymbol{\tau} + \mathbf{C}^T(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(\mathbf{q}) - \mathbf{F}_f(\dot{\mathbf{q}}) + \mathbf{Q}_{\text{ext}}$$

We define the **Generalized Momentum Observer Residual Vector $\mathbf{r}(t)$**:

$$\mathbf{r}(t) = \mathbf{K}_O \left[\mathbf{p}(t) - \int_0^t \left(\boldsymbol{\tau} + \mathbf{C}^T \dot{\mathbf{q}} - \mathbf{G} - \mathbf{F}_f + \mathbf{r} \right) d\sigma - \mathbf{p}(0) \right]$$

where $\mathbf{K}_O = \text{diag}(K_{O,1}, K_{O,2}) > 0$ is the observer diagonal gain matrix.

Differentiating $\mathbf{r}(t)$ with respect to time yields the decoupled error dynamics:

$$\dot{\mathbf{r}}(t) = \mathbf{K}_O \left(\dot{\mathbf{p}} - \left[\boldsymbol{\tau} + \mathbf{C}^T \dot{\mathbf{q}} - \mathbf{G} - \mathbf{F}_f + \mathbf{r} \right] \right) = \mathbf{K}_O \left(\mathbf{Q}_{\text{ext}} - \mathbf{r} \right)$$

In the Laplace domain, each channel acts as an independent first-order low-pass filter:

$$\frac{r_i(s)}{Q_{\text{ext},i}(s)} = \frac{K_{O,i}}{s + K_{O,i}}$$

$$\lim_{t \to \infty} \mathbf{r}(t) = \mathbf{Q}_{\text{ext}}$$

Notice: The observer estimates $\mathbf{Q}_{\text{ext}}$ using ONLY joint position $\mathbf{q}$, velocity $\dot{\mathbf{q}}$, and motor torque $\boldsymbol{\tau}$. Not a single acceleration term $\ddot{\mathbf{q}}$ appears anywhere in the algorithm!

6.3 Discrete Implementation & The 3-State Kalman-Filtered GMO (KF-GMO)

In Milestone 3 (`M3/scripts/build_M3_DiscreteObserver_30Hz.m`), this observer was discretized at the industrial PLC rate of **30 Hz** ($T_s = 1/30\,\text{s} \approx 33.3\,\text{ms}$) using exact First-Order Hold (FOH) matrix updates.

To eliminate filter phase lag under rapid impact loading and suppress resolver quantization noise, an upgraded **3-State Linear-Trend Kalman Filter** was integrated into the observer loop:

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{p}_k \\ \boldsymbol{\tau}_{L,k} \\ \dot{\tau}_{L,k} \end{bmatrix}$$

with state-dependent measurement covariance:

$$\mathbf{R}_k(d_4) = \left(D_{11}(d_4) \cdot \sigma_v\right)^2$$

Verified M3 Performance Audit (from `claims.md`):

- **Noise Suppression:** Resolver velocity noise was reduced from ** $193.15\,\text{kN}\cdot\text{m}$ to $36.10\,\text{kN}\cdot\text{m}$ (an 81.3% noise reduction!)**.

- **Lag Elimination:** Harmonic tracking lag was reduced from ** $120.8\,\mathrm{kN \cdot m}$ down to $16.26\,\mathrm{kN \cdot m}$**.
- **Friction Coverage:** Strict 100.0% coverage under Coulomb/viscous boundary conditions.
- **Zero-Load Noise Floor:** Reached **$1.35\,\mathrm{N \cdot m}$** (KF-GMO), easily distinguishing external rock resistance from internal powertrain friction.

---

7. Electromechanical Drive Dynamics, Wire Ropes & Rotor Inertia Reflection

In heavy mining machinery, electric motors do not connect directly to the dipper; they transmit power through high-ratio gearboxes, grooved drums, and flexible wire ropes.

7.1 Quadratic Rotor Inertia Reflection: The G_h^2 Amplification

In the TZ WK-20, two $600\,\mathrm{kW}$ AC induction motors drive the hoist drum through a reduction ratio of $G_h = 31.5 : 1$.

The total equivalent rotational inertia $J_{\mathrm{eq},h}$ reflected to the hoist drum shaft is:

$$J_{\mathrm{eq},h} = J_D + G_h^2 \eta_h J_{m,\mathrm{tot}}$$

where:

- $J_D = 4,200\,\mathrm{kg \cdot m^2}$ (Drum inertia).
- $J_{m,\mathrm{tot}} = 2 \times 28.5 = 57.0\,\mathrm{kg \cdot m^2}$ (Motor rotor inertia).
- $G_h = 31.5 \implies G_h^2 = 992.25$.
- $\eta_h = 0.93$ (Gearbox forward efficiency).

Carrying out the multiplication:

$$J_{m,\text{reflected}} = (992.25)(0.93)(57.0) = \mathbf{52,600.8}\,\text{kg}\cdot\text{m}^\mathbf{2}$$

$$J_{\text{eq},h} = 4,200 + 52,600.8 = \mathbf{56,800.8}\,\text{kg}\cdot\text{m}^\mathbf{2}$$

> **WARNING**
>
> ### WHY ROTOR INERTIA CAN NEVER BE NEGLECTED
>
> The high-speed motor rotors represent **92.6% of the total rotational inertia of the entire hoist drive!**
>
> If an observer model neglects rotor inertia, the immense acceleration torque required during rapid hoist reversals will be mistakenly identified by the observer as a massive rock collision on the dipper teeth, generating false emergency overload trips.

7.2 Wire Rope Elasticity & Unilateral Slackness

The twin $64\,\text{mm}$ hoist ropes on the TZ WK-20 have an effective modulus $E_{\text{rope}} = 115\,\text{GPa}$ (due to the helical lay of IWRC construction) and total metallic cross-sectional area $A_{\text{metal}} = 3,860.4\,\text{mm}^2$.

The axial stiffness varies inversely with suspended rope length $L_h(t)$:

$$k_r(L_h) = \frac{E_{\text{rope}} A_{\text{metal}}}{L_h(t)} \quad \left[\frac{\text{N}}{\text{m}}\right]$$

At $L_h = 14.5\,\text{m}$, $k_r = 30.62\,\text{MN/m}$. Under maximum digging stall tension ($1.65\,\text{MN}$), the elastic stretch is **$\Delta L_h = 53.9\,\text{mm}$** (exceeding $118\,\text{mm}$ under peak shock loading).

The wire rope cannot push. The physical tension T_h satisfies the unilateral complementarity condition:

$$T_h = \begin{cases} k_r \delta_c + c_r \dot{\delta}_c, & \text{if } \delta_c > 0 \text{ and } (k_r \delta_c + c_r \dot{\delta}_c) > 0 \\ 0, & \text{otherwise (Cable Slack)} \end{cases}$$

where $\delta_c = L_{h,\text{geom}}(\mathbf{q}) - L_{h,\text{drum}}(\theta_{\text{drum}})$. In ODE solvers, this is regularized using a smooth hyperbolic tangent function to avoid zero-crossing solver chattering.

7.3 Hoist-Crowd Involuntary Retraction Coupling

By the Principle of Virtual Work, the generalized actuator force vector $\boldsymbol{\tau}_{\text{act}}$ projected into configuration space is:

$$\boldsymbol{\tau}_{\text{act}} = \begin{bmatrix} \tau_{q_3} \\ \tau_{d_4} \end{bmatrix} = \begin{bmatrix} -h_h(\mathbf{q}) T_h \\ F_{\text{crowd}} - T_h \cos \beta_h(\mathbf{q}) \end{bmatrix}$$

Because the hoist cable pulls at an acute/obtuse angle relative to the handle, **hoisting pulls the dipper handle backward toward the shipper shaft with force $T_h \cos \beta_h$**.

Under heavy digging ($T_h = 620\,\mathrm{kN}$), this involuntary retracting force reaches **$549.3\,\mathrm{kN}$. **The crowd motor must output over** $5\,\mathrm{kN \cdot m}$ of shaft torque purely to hold the handle stationary against the hoist pull**!

---

8. Summary of Theoretical Chapters Authored (Volumes I & II)

To date, **7 complete, publication-grade research chapters (327 pages total)** have been written, peer-compiled, and typeset in large-format notebook style:

Chapter	Full Research Chapter Title	Pages	V
Chapter 1	Advanced Vector Mathematics, Linear Spaces & Matrix Calculus	36	P
Chapter 2	Multivariable Analysis, Kinematic Jacobians & Diff. Geometry	53	P
Chapter 3	Variational Principles, Generalized Coords & Euler-Lagrange	58	P
Chapter 4	Linear Parameter-Separable Regressors & System Identification	53	
Chapter 5	Advanced Nonlinear Observers & Momentum Formulations (M3)	41	P
Chapter 6	Top-Down Kinematic & Topological Decomposition of Shovel	34	P
Chapter 7	Planar 2-DOF Crowd-Hoist Dynamics & Cable-Driven Kinematics	52	
TOTAL	COMPLETE RESEARCH NOTEBOOK CURRICULUM TO DATE	327	A

All 7 chapters are accessible directly in the `study_notebook/` folder with complete clickable links:

- `Volume_01_Chapter_01_LargeNotebook.pdf`
- `Volume_01_Chapter_02.pdf`
- `Volume_01_Chapter_03.pdf`
- `Volume_01_Chapter_04.pdf`
- `Volume_01_Chapter_05.pdf`
- `Volume_02_Chapter_06.pdf`

- `Volume_02_Chapter_07.pdf`

9. Repository Codebase Guide: How to Run the Verified Models

Every theoretical claim in this project is tied to executable code in the repository. Here is the operational file guide:

```
c:\Users\IIT KGP\MATLAB Drive\shovel-dynamics\
│
├── claims.md                          <-- MASTER TRUTH DICTIONARY: Every number & so
├── research_log.md                    <-- CHRONOLOGICAL EXPERIMENT LOG: Dates, runs,
│
├── M1/                                <-- MILESTONE 1: Dynamic Model & Baseline Iden
│   ├── scripts/
│   │   ├── shovel_params_rasuli.m     <-- P&H 2100BL baseline parameters
│   │   ├── shovel_params_wk20.m       <-- TZ WK-20 Amlohri parameters
│   │   ├── forward_dynamics.m         <-- M1 forward simulation engine
│   │   └── inverse_dynamics.m         <-- M1 inverse torque verification
│   └── experiments/                   <-- Stage 1 blind identification verification
│
├── M2/                                <-- MILESTONE 2: External Load Recovery (MS02P
│   ├── 00_design/                     <-- Frozen experimental specification
│   ├── experiments/
│   │   ├── 01_zero_force/             <-- MS02P01: Numerical floor test
│   │   ├── 02_known_load/             <-- MS02P02: 100 kN vertical recovery
│   │   ├── 03_amplitude_sweep/        <-- MS02P03: 10-1000 kN linearity
│   │   ├── 04_frequency_sweep/        <-- MS02P04: Trajectory bandwidth test
│   │   ├── 05_configuration_sweep/    <-- MS02P05: Crowd bias reach test
│   │   ├── 06_identifiability/        <-- MS02P06: Formal SVD & collinearity scripts
│   │   ├── 07_model_perturbation/     <-- MS02P07: Geometric error attribution
│   │   └── 08_held_out/               <-- MS02P08: Torque-driven dynamic deflection
│   └── scripts/
│       ├── build_M2_TorqueDriven.m    <-- Full closed-loop forward dynamic simulator
│       └── run_M2_P06_FormalSVD.m     <-- SVD collinearity & subspace angle calculat
│
├── M3/                                <-- MILESTONE 3: Momentum Observers & Kalman F
│   ├── scripts/
│   │   └── build_M3_DiscreteObserver_30Hz.m <-- Verified 30 Hz Simulink discrete KF-
│   └── 06_RESULTS_MAT/                <-- Verified .mat archives (noise floor 1.35 N
│
└── study_notebook/                    <-- AUTHORED VOLUMES (PDF, HTML, DOCX, MD)
```

How to Run Key Verification Scripts in MATLAB:

1. **To verify the M2 SVD Identifiability & Collinearity (MS02P06):**

```matlab
cd 'c:\Users\IIT KGP\MATLAB Drive\shovel-dynamics\M2\experiments\06_identifiability'
run('run_M2_P06_FormalSVD.m')
% Outputs: Single-channel kappa = 497.71 vs. Joint 2-channel kappa = 3.77
```

1. **To run the forward torque-driven dynamic deflection simulation (MS02P08):**

```matlab
cd 'c:\Users\IIT KGP\MATLAB Drive\shovel-dynamics\M2\scripts'
run('build_M2_TorqueDriven.m')
% Simulates closed-loop dynamic deflection and 30 Hz observer tracking
```

1. **To verify the native 30 Hz Discrete Momentum Observer & KF-GMO (M3):**

```matlab
cd 'c:\Users\IIT KGP\MATLAB Drive\shovel-dynamics\M3\scripts'
run('build_M3_DiscreteObserver_30Hz.m')
% Verifies 81.3% noise reduction and 1.35 Nm noise floor
```

---

10. Doctoral Oral Defense: The 10 Critical Viva Questions & Answers

If you are asked to defend this research before an academic committee or industrial evaluation board, use these exact, authoritative answers:

Q1: Why can you not simply use load cells on the bail pin or teeth to measure digging forces?

Answer:

In opencast mining, the dipper teeth endure explosive cyclic impacts exceeding $1.5\,\mathrm{MN}$ amidst severe abrasive rock slurry and extreme shock vibrations ($> 100\,\mathrm{g}$). Physical strain gauges or load cells mounted near the cutting edge fail within days due to mechanical abrasion, lead wire severing, or connector destruction. Our physics-based digital twin estimates external interaction forces **completely sensorlessly**, utilizing existing, ruggedized motor resolvers and inverter drive currents located safely inside the protected machinery house.

Q2: What is the exact mathematical difference between payload estimation and external load estimation?

Answer:

Payload is a **mass change** (m_{pay}). It modifies the kinetic energy, entering the equations of motion as an augmentation of the mass matrix $\mathbf{D}(\mathbf{q})$, the Coriolis matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$, and the gravity vector $\mathbf{G}(\mathbf{q})$. An external load $\mathbf{F}_{\mathrm{ext}}$ is a **non-conservative generalized force vector** $\mathbf{Q}_{\mathrm{ext}} = \mathbf{J}_T^T \mathbf{F}_{\mathrm{ext}}$ entering purely on the right-hand side of the manipulator equation. Milestone 2 validates the

recovery of the generalized force vector $\mathbf{Q}_{\text{ext}}$; payload mass estimation is performed during the steady swing phase where $\mathbf{Q}_{\text{ext}} = \mathbf{0}$.

Q3: Why does single-channel saddle torque fail to separate payload from dipper mass, and how did you resolve it?

Answer:

In MS02P06, we proved through Singular Value Decomposition (SVD) that in the pitch equation alone, the regressor columns for dipper mass $M_d g(d_4 + c)\cos q_3$ and external vertical load $F_y(d_4 + L_{bx})\cos q_3$ are nearly identical, with a correlation coefficient $r = 0.9999919$, condition number $\kappa_2 = 497.71$, and subspace angle $\theta_{\min} = 0.23^\circ$. They are mathematically collinear. We resolved this excitation trilemma by formulating a **joint 2-channel estimator** combining saddle pitch torque and crowd rack thrust scaled by a characteristic dimension $\bar{s} = 10.0\,\text{m}$. This expanded the subspace angle to $\theta_{\min} = 29.70^\circ$ and dropped the condition number to $\kappa_2 = 3.77$, rigorously guaranteeing independent parameter separability.

Q4: Why must we use a Generalized Momentum Observer instead of standard inverse dynamics $\boldsymbol{\tau} - \mathbf{D}\ddot{\mathbf{q}}$?

Answer:

Standard inverse dynamics requires double differentiation of encoder positions to obtain joint accelerations $\ddot{\mathbf{q}}$. In real machinery, encoder quantization and electrical inverter noise are amplified by $\frac{1}{\Delta t^2}$. At 30 Hz, this noise amplification factor is 900, completely corrupting the acceleration signal. The Generalized Momentum Observer utilizes the Cardinal Momentum Identity $\dot{\mathbf{D}}\dot{\mathbf{q}} - \mathbf{C}\dot{\mathbf{q}} = \mathbf{C}^T\dot{\mathbf{q}}$ to express system momentum dynamics purely as an integral of position, velocity, and motor torque, **completely bypassing the need for acceleration sensing**.

Q5: How does the 3-state KF-GMO handle resolver quantization noise and phase lag?

Answer:

Standard low-pass filters introduce fatal time lag ($50\,\text{ms} - 120\,\text{ms}$), causing severe phase error during rapid rock impact. In M3, we implemented a 3-state linear-trend Kalman Filter ($[\mathbf{p}; \boldsymbol{\tau}_L; \dot{\tau}_L]$) with state-dependent covariance $\mathbf{R}_k(d_4) = (D_{11}(d_4)\sigma_v)^2$. This observer tracks acceleration trends while suppressing high-frequency velocity variance by **81.3% (from $193.15\,\text{kN}\cdot\text{m}$ down to $36.10\,\text{kN}\cdot\text{m}$) **and slashing tracking lag from** $120.8\,\text{kN}\cdot\text{m}$ to $16.26\,\text{kN}\cdot\text{m}$**.

Q6: Why does the hoist wire rope tension induce an involuntary retraction force on the crowd drive?

Answer:

Because the hoist rope line connecting the boom point sheave S to bail pin B does not remain perpendicular to the dipper handle; it forms an angle β_h with the handle centerline. The axial projection of cable tension along the handle is $-T_h \cos\beta_h = -T_h \frac{\partial L_h}{\partial d_4}$. Under heavy hoisting ($620\,\text{kN}$), this pulls the handle backward into the saddle with over $549\,\text{kN}$ of force. The crowd motor must output forward torque purely to counter this cable retraction force.

Q7: Why is the hoist drum equivalent inertia dominated by the motor rotor rather than the massive steel drum?

Answer:

Because rotational inertia reflected across a gear reducer scales with the **square of the gear ratio** (G_h^2). In the TZ WK-20 hoist drive, $G_h = 31.5$, so $G_h^2 = 992.25$. Even though the twin motor rotors weigh only $57\,\text{kg}\cdot\text{m}^2$,

their reflected inertia at the drum shaft is $52,600.8\,\mathrm{kg}\cdot\mathrm{m}^2$, accounting for ** 92.6% of the total drive inertia**.

Q8: What was the finding regarding the geometric offset c in MS02P07?

Answer:

We evaluated tolerance perturbation and discovered that a small $+0.10\,\mathrm{m}$ error ($+7.6\%$) in the physical center-of-mass lever arm c couples directly into the dynamic regressor, producing a **$+756.6\,\mathrm{kg}$ false mass misattribution**. This proves that mechanical dimensions cannot be assumed; they must be accurately calibrated via laser metrology or blind geometric identification prior to estimating payloads.

Q9: Why can you NOT transfer P&H 2100BL parameters to the TZ WK-20 digital twin?

Answer:

The P&H 2100BL is a $350\,\mathrm{t}$ machine with an $18.30\,\mathrm{m}$ boom and symmetric shipper shaft location ($L_{b1} = L_{b2} = 9.15\,\mathrm{m}$). The TZ WK-20 is an $850\,\mathrm{t}$ machine with a $21.50\,\mathrm{m}$ boom and asymmetric shipper shaft location ($L_{b1} = 10.25\,\mathrm{m}, L_{b2} = 11.25\,\mathrm{m}$). Transferring P&H geometry causes an error of over $2.1\,\mathrm{m}$ in sheave distance and shifts the hoist moment arm h_h by up to 22%, completely corrupting force estimation.

Q10: How does this digital twin directly support Prognostics and Health Management (PHM)?

Answer:

By separating baseline dynamic torques (inertia, Coriolis, gravity, nominal friction) from real-time motor signals, the digital twin generates **clean physical**

residuals:

1. **Friction Drift Residual:** Detects rack-and-pinion wear and bushing lubrication starvation before mechanical seizure.
2. **External Stress History:** Feeds reconstructed digging impact loads into Rainflow cycle counting and Paris-Erdogan crack propagation models to predict the **Remaining Useful Life (RUL)** of the boom structure.
3. **Electrical Degradation:** Compares commanded electromagnetic torque $\tau_e = K_t I_q$ against mechanical reaction torque to detect early rotor bar cracking or stator insulation breakdown in the 690 V AC drives.

Executive Sign-Off & Recommendation

This master synthesis document reflects the complete, verified, and audited state of the project. It integrates every result from **M1 (model), M2 (load recovery), M3 (momentum observer & Kalman filter), Chapters 1 to 7 (foundations & kinematics), and the Machine A / Machine B parameter databases**.

You can now use this document as your primary research guide, presentation basis, and thesis foundation.